CISS350: Data Structures and Advanced Algorithms
Quiz q10401

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Q1. You want to sort an array of size n (which is ≥ 1000000000) of integer value in the range 0..9. You want the fastest sorting algorithm. Complete the following function to essentially perform a counting sort on `x`.
Answer:

```
// Sort x[0..n-1]
void sort(int x[], int n)
{
  int count[10] = {0};

  for(int i = 0; i < n; ++i)
  {
    count[x[i]]++;
  }

  int index = 0;
  for(int i = 0; i < 10; i++)
  {
    while(count[i] > 0)
    {
      x[index++];
      count[i]--;
    }
  }
}
```

Q2. You want to sort an array of size n (which is ≥ 1000000000). of integer value in the range 100000..100020. The device is memory-constrained – besides the input array, you can afford to use at most the equivalent of 100 integers for your computations (for variables). Complete the following function to essentially perform a counting sort on `x`.
Answer:

```
// Sort x[0..n-1]
void sort(int x[], int n)
{
  const int min = 100000;
  const int range = 21;

  count[range] = 0;
```

```
  for (int i = 0; i < n; ++i)
  {
    count[x[i] - MIN]++;
  }

  int index = 0;
  for (int i = 0; i < RANGE; ++i)
  {
    while (count[i] > 0)
    {
      x[index++] = i + MIN;
      count[i]--;
    }
  }
}
```

Q3. You want to sort an array of size n (which is ≥ 1000000000) and contains double values -5.0, -4.5, -4.0, -3.5, -3.0, ..., -0.5, 0.0, 0.5, ..., 4.5, 5.0. The device is memory-constrained – besides the input array, you can afford to use at most the equivalent of 100 integers for your computations (for variables). Complete the following function to essentially perform a counting sort on `x`.

Answer:

```
// Sort x[0..n-1]
void sort(double x[], int n) // correction: int x[] should be double x[].
{
  const int RANGE = 21; // -5.0 to 5.0 step 0.5
  int count[RANGE] = {0};

  for (int i = 0; i < n; ++i)
  {
    int idx = (int)((x[i] + 5.0) / 0.5);
    count[idx]++;
  }

  int index = 0;
  for (int i = 0; i < RANGE; ++i)
  {
    double value = -5.0 + i * 0.5;
    while (count[i] > 0)
    {
      x[index++] = value;
      count[i]--;
    }
  }
}
```

Q4. Given a string, you want to return the character which appears most frequently in

the string. For instance if the most frequently occurring character in `"hello world"` is `'l'`. Use the idea from `countingsort` to count the number of times a character occurs in a string `s` and then perform a max computation to compute the most commonly occurring character in the given string.

Answer:

```
char most_common_char(char s[])
{
  int count[256]; // count[i] is the number of times char(i) occurs in s

  for (int i = 0; s[i] != '\0'; ++i)
  {
    count[(int)s[i]]++;
  }

  int max = 0;
  char result = s[0];

  for (int i = 0; i < 256; ++i)
  {
    if (count[i] > max)
    {
      max = count[i];
      result = (char)i;
    }
  }

  return result;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.